Computational Statistical Catallactics

Brief for Jake Gunther & Marc Dotson — Reframing Agent-Based Economics on Mises–Hayek–de Finetti Microfoundations

TJB

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Re: Bayesians in the Catallaxy: From Mises–Hayek to de Finetti

Framing

The paper's methodological proposal is computational statistical catallactics (CSC) — a research program whose microfoundations are Mises, Hayek, and de Finetti, whose method is computational simulation, and whose central claim is that stochasticity in market outcomes is endogenous: generated by purposive individual action under genuine uncertainty rather than imposed from outside by an objective probability distribution.

The methodology section argues that agent-based computational economics (ACE) has been "a method in search of a theory" and that catallactics has been "a theory in search of a method"; CSC supplies each with what the other lacks. To make this case persuasively to a JEP audience, we need to reframe the existing ACE and Bayesian literature as already on the way to CSC — work whose practitioners have been reaching for principled microfoundations they have been unable to articulate from within their own framing.

TJB will share a curated reading list. Your job is to read, summarize, and synthesize it into a briefing document — the synthesis itself is the deliverable. TJB will use the briefing as source material for section 4 of the manuscript and the corresponding section of the short proposal.

The tasks

A. Read and summarize the literature TJB sends

- One to two paragraphs per major item, plus a synthesis.
- For each piece: what it argues, where it succeeds, and what it cannot articulate from within its own framing.
- The integration target: identify, in each paper, the implicit move toward Mises–Hayek–de Finetti microfoundations, and the place at which the lack of those foundations becomes a binding methodological constraint.

B. Reframe ACE as "now existing"

- ACE is a method that has been around for decades. The paper's claim is that CSC is not a new method but the theoretical reframing under which existing ACE work becomes properly grounded.
- The section should acknowledge ACE's accomplishments generously and show, with examples from the literature, where the lack of principled subjective-probability foundations leaves ACE models methodologically ad hoc — agents given probability distributions and learning rules from outside, with "uncertainty" reduced to parametric risk.
- The framing is cooperative, not adversarial. CSC is the theoretical home that ACE has been waiting for.

C. Bayesian agent-based models *(natural anchor for Marc)*

- Marc's specialty is Bayesian modeling in marketing. Independently, he and TJB share a developing interest in the links between marketing and financial market microstructure — the crossover is something we are looking to explore together. For this brief, the contribution comes in two pieces:
  - Established ground. Where in the Bayesian or agent-based literature have models already begun to do (informally) what CSC proposes (formally)? Where is the marketing-side use of Bayesian agents reaching for principled subjective-probability foundations and not quite finding them?
  - Lines of contact. Where do you see promising bridges from Bayesian marketing models toward the market-microstructure / catallactic story we are building? These are exploratory; we expect to develop them together rather than fully resolve them in this brief.

D. Computational and signals-processing infrastructure *(natural anchor for Jake)*

- Jake's signals-processing perspective is the right lens on the practical computational side of CSC.
- What does the simulation/estimation infrastructure look like? Section 4 sketches options-market market-makers as the proof-of-concept application; Jake's expertise should help articulate what data and computational primitives that requires, and what existing technology (signal processing on order-book data, electronic limit-order-book microstructure simulation, high-frequency data infrastructure) the program can build on.
- What are the important questions, issues, and methods from computational statistics and the Monte Carlo literature that are important for CSC?

E. Strengthen the options-markets proof-of-concept

- Section 4 of the manuscript already proposes options-market-makers as Misesian entrepreneurs in a CSC framework, building on Figlewski.

- Strengthen this with appropriate citations from the agent-based and Brock-style statistical finance literature, and where promising, draw on the Bayesian marketing literature for parallels in modeling subjective belief and choice.

What we are not asking

- Reframe or reopen the economic argument from sections 1–3 (the de Finetti / communism-of-models / exchangeable-search insights).
- Write CSC as a competitor or successor methodology to ACE in adversarial terms. The right framing is that CSC is the theoretical home that ACE has been waiting for.

Suggested deliverable

A 5–8 page briefing document organized around tasks A–E, with the synthesis of the readings as its core. This will become source material for section 4 of the manuscript and the corresponding section of the short proposal; we are not asking you to draft those sections directly.

Logistics

- TJB will send the curated reading list separately.
- The full current manuscript is in `chunks/01-introduction.qmd` through `chunks/04-computational-statistical-catallactics.qmd`. The compact statement of the argument is in `proposal/proposal-short.qmd`. Read both before starting.